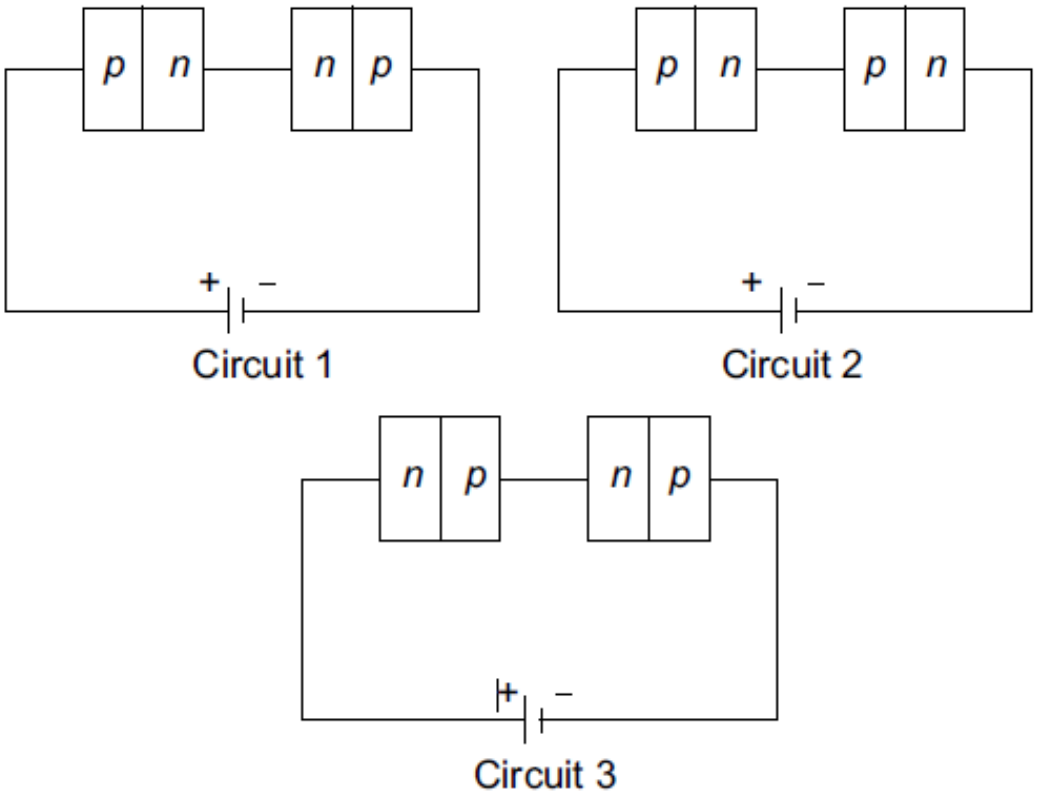
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Two identical $p-n$ junctions may be connected in series with a battery in three ways (Fig.). The potential drops across the two $p-n$ junctions are equal in



(A) circuit 1 and circuit 2

(B) circuit 2 and circuit 3

(C) circuit 3 and circuit 1

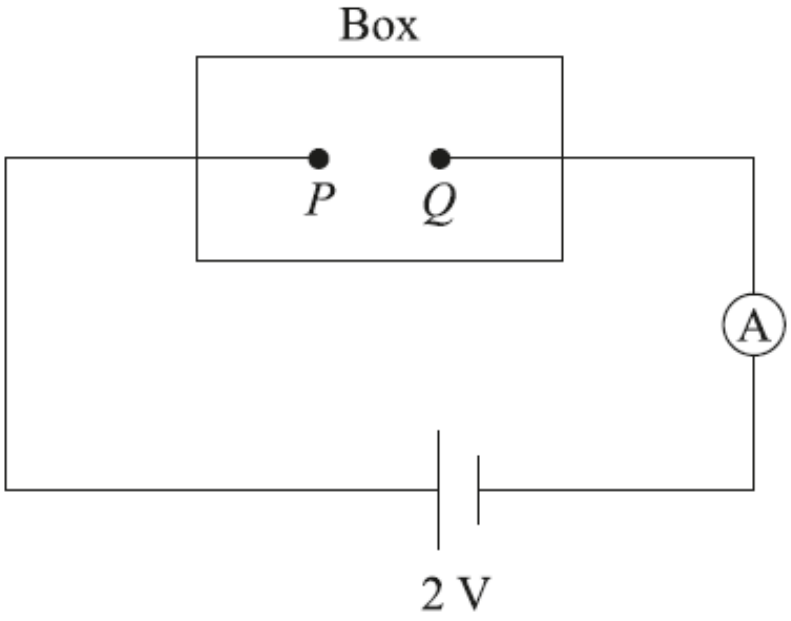
(D) circuit 1 only

Q.2

A junction diode and resistor are connected together between terminals P and Q in a box. This combination is connected to a battery of voltage $2\ \mathrm{V}$ and a milliammeter as shown in Fig. 20.56. The current in the circuit is $40\ \mathrm{mA}$. When the terminals of the battery are interchanged, the current drops to $20\ \mathrm{mA}$. The resistances of

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the diode and resistor are



(A) $100\Omega, 100\Omega$

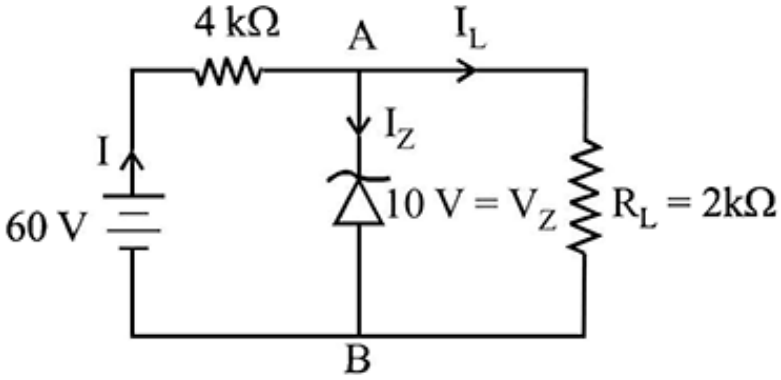
(B) $50\Omega, 100\Omega$

(C) $50\Omega, 50\Omega$

(D) $50\Omega, 150\Omega$

Q.3

A Zener diode is connected to a battery and a load as show below:
The currents, I, I $_Z$ and I_L are respectively.



(A) $15\ \text{mA}, 5\ \text{mA}, 10\ \text{mA}$

(B) $15\ \text{mA}, 7.5\ \text{mA}, 7.5\ \text{mA}$

(C) $12.5\ \text{mA}, 5\ \text{mA}, 7.5\ \text{mA}$

(D) $12.5\ \text{mA}, 7.5\ \text{mA}, 5\ \text{mA}$

Q.4

A sinusoidal voltage of amplitude 25 volt and frequency $50\ \text{Hz}$ is applied to a half wave rectifier using P-n junction diode. No filter is used and the load resistance is 1000Ω. The forward resistance R_f of ideal diode is 10Ω. The percentage efficiency of rectifier is

(A) 40%

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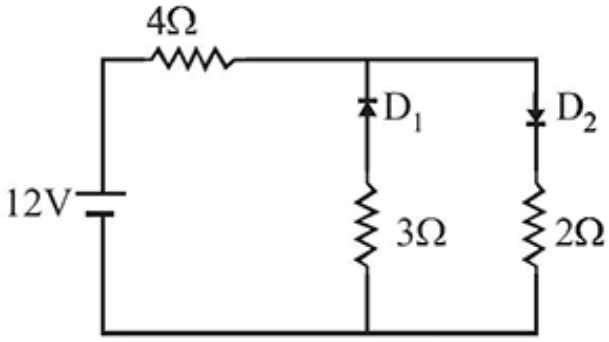
(B) 20%

(C) 30%

(D) 15%

Q.5

The circuit has two oppositively connected ideal diodes in parallel. The current flowing in the circuit is



(A) 1.71 A

(B) 2.00 A

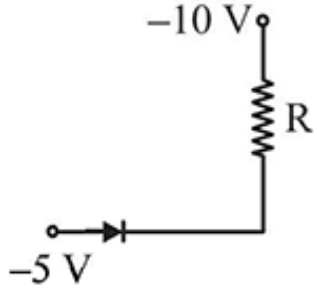
(C) 2.31 A

(D) 1.33 A

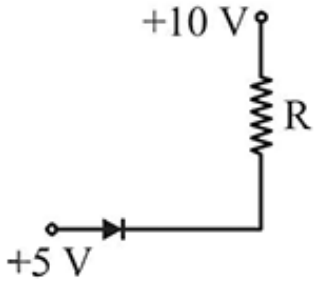
Q.6

Which of the junction diodes shown below are forward biased?

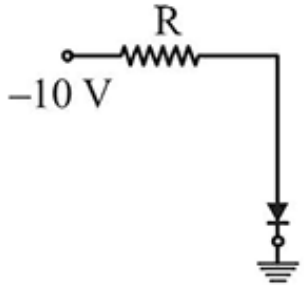
(A)



(B)

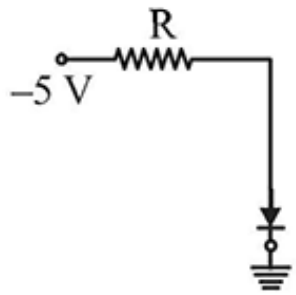


(C)



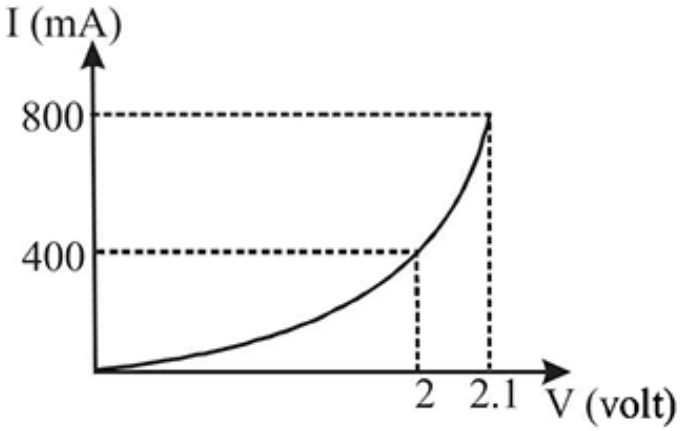
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(D)



Q.7

The I-V characteristic of a P-N junction diode is shown below. The approximate dynamic resistance of the p-n junction when a forward bias voltage of 2 volt is applied is



(A) 1Ω

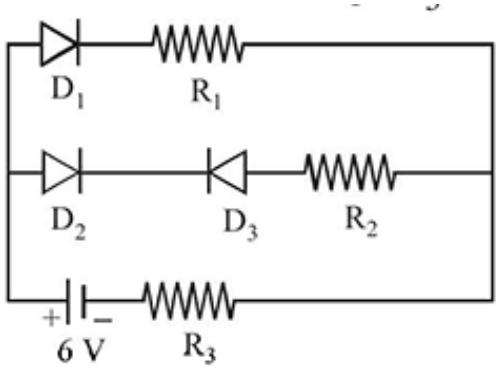
(B) 0.25Ω

(C) 0.5Ω

(D) 5Ω

Q.8

Figure shows a circuit in which three identical diodes are used. Each diode has forward resistance of 20Ω and infinite backward resistance. Resistors $R_1 = R_2 = R_3 = 50\Omega$. Battery voltage is $6\ V$. The current through R_3 is:



(A) $50\ mA$

(B) $100\ mA$

(C) $60\ mA$

(D) $25\ mA$

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Q.9

When the forward bias voltage of a diode is changed from $0.6\ \mathrm{V}$ to $0.7\ \mathrm{V}$, the current changes from $5\ \mathrm{mA}$ to $15\ \mathrm{mA}$. Then its forward bias resistance is

(A) 0.01Ω

(B) 0.1Ω

(C) 10Ω

(D) 100Ω

Q.10

Zener breakdown in a semi-conductor diode occurs when

(A) forward current exceeds certain value

(B) reverse bias exceeds certain value

(C) forward bias is equal to certain value

(D) potential barrier is reduced to zero

Q.11

A silicon diode has a threshold voltage of $0.7\ \mathrm{V}$. If an input voltage given by $2\sin(\pi t)$ is supplied to a half wave rectifier circuit using this diode, the rectified output has a peak value of

(A) $2\ \mathrm{V}$

(B) $1.4\ \mathrm{V}$

(C) $1.3\ \mathrm{V}$

(D) $0.7\ \mathrm{V}$

Q.12

A potential barrier of $0.50\ \mathrm{V}$ exists across a $\mathrm{p}-\mathrm{n}$ junction. If the depletion region is $5.0\times10^{-7}\ \mathrm{m}$ wide, the intensity of the electric field in this region is

(A) $1.0\times10^{6}\ \mathrm{V/m}$

(B) $1.0\times10^{5}\ \mathrm{V/m}$

(C) $2.0\times10^{5}\ \mathrm{V/m}$

(D) $2.0\times10^{6}\ \mathrm{V/m}$

Q.13

A zener diode of voltage $\mathrm{V_Z}(=6\ \mathrm{V})$ is used to maintain a constant voltage across a load resistance $\mathrm{R_L}(=$

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1000Ω) by using a series resistance R_c $(= 100\Omega)$. If the e.m.f. of source is $E(= 9\ V)$, what is the power being dissipated in Zener diode ?

(A) 0.144 watt

(B) 0.324 watt

(C) 0244 watt

(D) 0544 watt

Q.14

A diode having potential difference $0.5\ V$ across its junction which does not depend on current, is connected in series with resistance of 20Ω across source. If 0.1 A current passes through resistance then what is the voltage of the source?

(A) $1.5\ V$

(B) $2.0\ V$

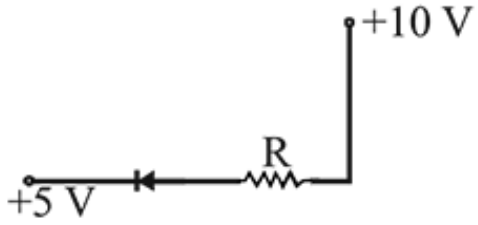
(C) $2.5\ V$

(D) $5\ V$

Q.15

In the following, which one of the diodes reverse biased?

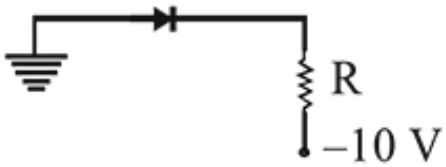
(A)



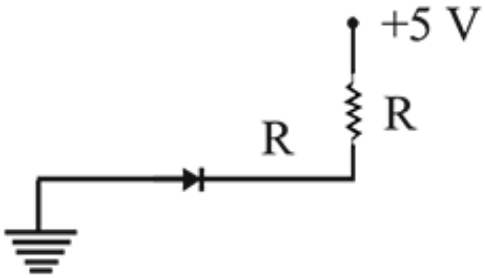
(B)



(C)



(D)



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Q.16

Suppose a 'n'-type wafer is creatd by doping Si crystal having 5×10^{28} atoms $/\mathrm{m}^3$ with 1ppm concentration of As. On the surface 200 ppm Boron is added to create 'P' region in this wafer. Considering $n_i = 1.5 \times 10^{16}\ \mathrm{m}^{-3}$ calculate the density of the minority charge carriers in the p regions.

(A) $2.25 \times 10^7/\mathrm{m}^3$

(B) $1.12 \times 10^3/\mathrm{m}^3$

(C) $3.11 \times 10^6/\mathrm{m}^3$

(D) $2.11 \times 10^5/\mathrm{m}^3$

Q.17

Which of the following statements is incorrect?

(A) The drift current in a p-n junction is from the n -side to the p -side.

(B) Drift current is due to free electrons only.

(C) Silicon is preferred over germanium for making semiconductor devices.

(D) The energy gap in germanium is more than the energy gap in silicon.

Q.18

When a potential difference is applied across, the current passing through

(A) an insulator at $0\ \mathrm{K}$ is maximum

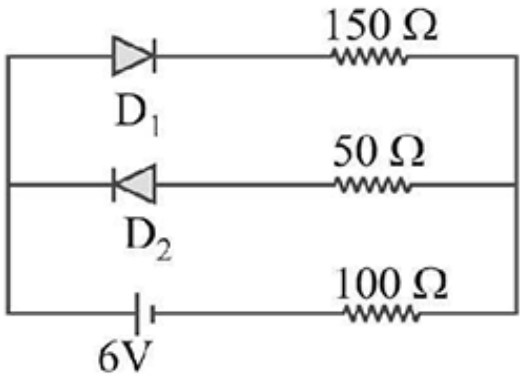
(B) a semiconductor at $0\ \mathrm{K}$ is zero

(C) a metal at $0\ \mathrm{K}$ is finite

(D) a P-N diodeat $300\ \mathrm{K}$ is zero, ifit is reverse biased

Q.19

The circuit shown in following figure contains two diode D_1 and D_2 each with a forward resistance of 50 ohms and with infinite backward resistance. If the battery voltage is $6\ \mathrm{V}$, the current through the 100 ohm resistance (in amperes) is



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(A) zero

(B) 0.02

(C) 0.03

(D) 0.036

Q.20

Consider an n-p-n transistor amplifier in common emitter configuration. The current gain of the transistor is 100 . If the collector current changes by $1\ \mathrm{mA}$, what will be the change in emitter current

(A) $1.1\ \mathrm{mA}$

(B) $1.01\ \mathrm{mA}$

(C) $0.01\ \mathrm{mA}$

(D) $10\ \mathrm{mA}$

Answers & Solutions

Q.1 Answer:

circuit 2 and circuit 3

Solution:

The potential drops are equal when the resistances offered by the two $p-n$ junctions are equal. In circuit 1 the first junction diode is forward biased having low resistance whereas the second junction diode is reversed biased having a very high resistance. Hence V_1, the potential drop across the first $p-n$ junction is very low as compared to the second $p-n$ junction.
In circuit 2 both junctions are forward biased having equal low resistances whereas in circuit 3 both junctions are reverse biased having equal high resistances. Hence the correct choice is (b).

Q.2 Answer:

$100\Omega, 100\Omega$

Solution:

Since the current decreases on interchanging the battery terminals, it is clear that the diode is forward biased in the first case and reverse biased in the second case. Furthermore, since the current does not drop to zero, it follows that the diode and the resistor are not connected in series because then the current should drop to zero. So they must be connected in parallel as shown in Fig. (a) and (b).
Let R_{d} be the resistance of the diode. In case (a) the diode is forward biased. Hence the current I_1 divides between R_{d} and R. The resistance of the parallel combination is
$R_{\mathrm{p}}=\frac{2\ \mathrm{V}}{40\times10^{-3}\ \mathrm{A}}=50\Omega$
In case (b), the diode is reverse biased and hence no current flow through it; all the current I_2 flows through R. The value of R is

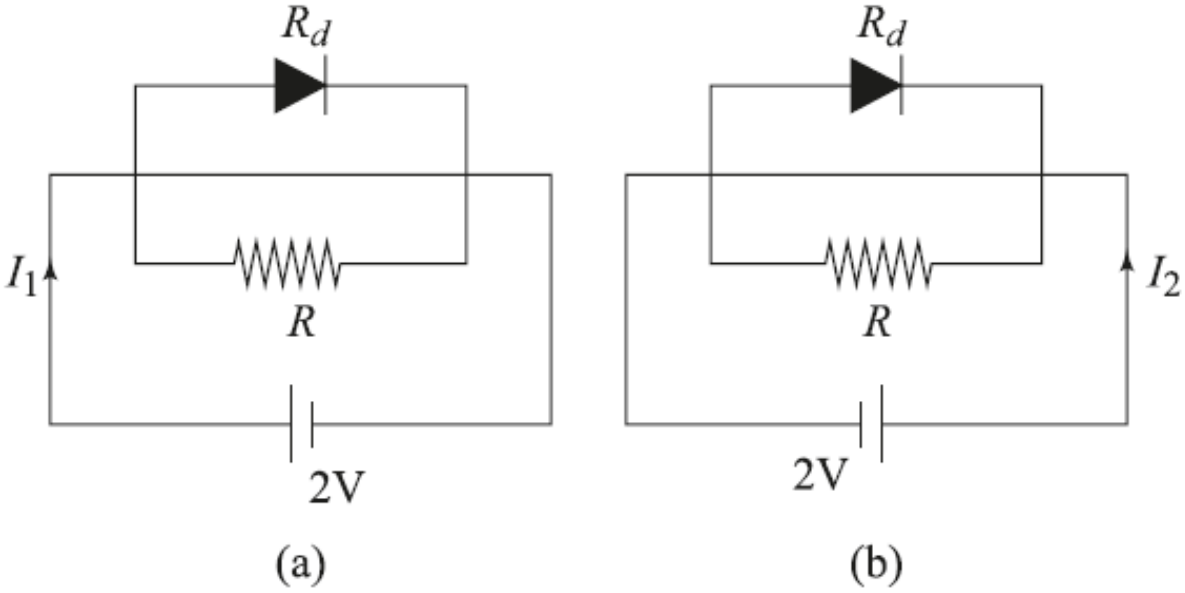
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$R = \frac{2\text{ V}}{20\times10^{-3}\text{ A}} = 100\Omega$

Now $\frac{1}{R_p} = \frac{1}{R} + \frac{1}{R_d}$

$\Rightarrow \quad \frac{1}{R_d} = \frac{1}{R_p} - \frac{1}{R} = \frac{1}{50} - \frac{1}{100} \Rightarrow R_d = 100\Omega$

So the correct choice is (a).



Q.3 Answer:

$12.5\text{ mA}, 7.5\text{ mA}, 5\text{ mA}$

Solution:

Here, $R = 4\text{k}\Omega = 4\times10^3\Omega$

$V_i = 60\text{ V}$

Zener voltage $V_z = 10\text{ V}\ R_L = 2\text{k}\Omega = 2\times10^3\Omega$

Load current, $I_\text{L} = \frac{V_Z}{R_L} = \frac{10}{2\times10^3} = 5\text{ mA}$

Current through $R, I = \frac{V_i - V_Z}{R}$

$= \frac{60-10}{4\times10^3} = \frac{50}{4\times10^3} = 12.5\text{ mA}$

From circuit diagram, $I = I_Z + I_L$

$\Rightarrow \quad 12.5 = I_Z + 5$

$\Rightarrow I_Z = 12.5 - 5 = 7.5\text{ mA}$

Q.4 Answer:

40%

Solution:

$I_\text{m} = \frac{\text{V}_\text{m}}{\text{R}_\text{f}+\text{R}_\text{L}} = \frac{25}{(10+1000)} = 24.75\text{ mA}$

$\text{I}_\text{dc} = \frac{\text{I}_\text{m}}{\pi} = \frac{24.75}{3.14} = 7.87\text{ mA}$

$\text{I}_\text{rms} = \frac{\text{I}_\text{m}}{2} = \frac{24.75}{2} = 12.37\text{ mA}$

$\text{P}_\text{dc} = \text{I}_\text{dc}^2 \times \text{R}_\text{L} = \left(7.87\times10^{-3}\right)^2 \times 10^3 = 61.9\text{ mW}$

$\text{P}_\text{ac} = \text{I}_\text{rms}^2\left(\text{R}_\text{f} + \text{R}_\text{L}\right) = \left(12.37\times10^{-3}\right)^2 \times (10 + 1000)$

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$= 154.54\ \text{mW}$

Rectifier efficiency $\eta = \frac{\mathrm{P_{dc}}}{\mathrm{P_{ac}}} \times 100 = \frac{61.9}{154.54} \times 100 = 40.05\%$

Q.5 Answer:

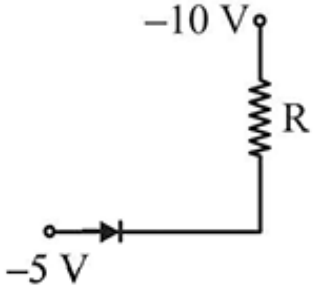
$2.00\ \text{A}$

Solution:

D_2 **is forward biased whereas** D_1 **is reversed biased. So effective resistance of the circuit** $R = 4 + 2 = 6\Omega$

$\therefore i = \frac{12}{6} = 2\ \text{A}.$

Q.6 Answer:



Solution:

Positive terminal is at higher potential $(-5\ \text{V})$ **and negative terminal is at lower potential** $-10\ \text{V}.$

Q.7 Answer:

0.25Ω

Solution:

$\mathrm{R} = \frac{\Delta \mathrm{V}}{\Delta \mathrm{I}} = \frac{2.1-2}{(800-400)\times 10^{-3}} = \frac{1}{4} = 0.25\Omega$

Q.8 Answer:

$50\ \text{mA}$

Solution:

Here, diodes $\mathrm{D_1}$ **and** $\mathrm{D_2}$ **are forward biased and** $\mathrm{D_3}$ **is reverse biased. Therefore current through** $\mathrm{R_3}$

$\mathrm{i} = \frac{\mathrm{V}}{\mathrm{R'}} = \frac{6}{120} = \frac{1}{20}\ \text{A} = 50\ \text{mA}$

Q.9 Answer:

10Ω

Solution:

$$\text{Forward bias resistance } = \frac{\Delta \mathrm{V}}{\Delta \mathrm{I}}$$
$$= \frac{(0.7-0.6)\mathrm{V}}{(15-5)\mathrm{mA}} = \frac{0.1}{10\times 10^{-3}} = 10\Omega.$$

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Q.10 Answer:

reverse bias exceeds certain value

Solution:

At the zener breakdown voltage the electric field of the junction is high $\left(10^{6}\ \mathrm{V/m}\right)$ enough to pull valence electrons from the host atoms. The emission of electrons from the host atoms accounts for high current at the breakdown

Q.11 Answer:

$1.3\ \mathrm{V}$

Solution:

Peak value of rectified output voltage = peak value of input voltage − barrier voltage $= 2 - 0.7 = 1.3\ \mathrm{V}$

Q.12 Answer:

$1.0 \times 10^{6}\ \mathrm{V/m}$

Solution:

$$\mathrm{E} = \frac{\mathrm{V}}{\mathrm{d}} = \frac{0.50}{5 \times 10^{-7}}\ \mathrm{V/m} = 1.0 \times 10^{6}\ \mathrm{V/m}$$

Q.13 Answer:

0.144 watt

Solution:

Here, $\mathrm{E} = 9\ \mathrm{V}; \mathrm{V_z} = 6; \mathrm{R_L} = 1000\Omega$ **and** $\mathrm{R_s} = 100\Omega$

Potential drop across series resistor $\mathrm{V} = \mathrm{E} - \mathrm{V_Z} = 9 - 6 = 3\ \mathrm{V}$

Current through series resistance $\mathrm{R_S}$ **is** $I = \frac{V}{R} = \frac{3}{100} = 0.03\ \mathrm{A}$

Current through load resistance $\mathrm{R_L}$ **is** $\mathrm{I_L} = \frac{\mathrm{V_Z}}{\mathrm{R_L}} = \frac{6}{1000} = 0.006\ \mathrm{A}$

Current through Zener diode is $\mathrm{I_Z} = \mathrm{I} - \mathrm{I_L} = 0.03 - 0.006 = 0.024\mathrm{amp}$

Power dissipated in Zener diode is $\mathrm{P_Z} = \mathrm{V_Z I_Z} = 6 \times 0.024 = 0.144\mathrm{Watt}$

Q.14 Answer:

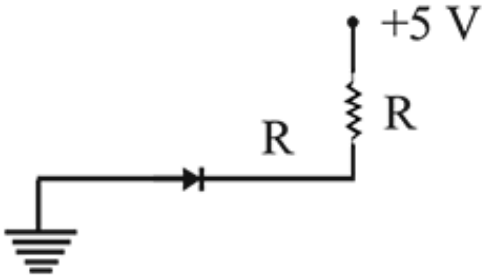
$2.5\ \mathrm{V}$

Solution:

No solution available

Q.15 Answer:

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Solution:

p **-side connected to low potential and** n **-side is connected to high potential.**

Q.16 Answer:

$2.25 \times 10^{7} / \mathrm{m}^{3}$

Solution:

In ' n **' region; number of** e^{-} **is due to As:**

$n_{e}=N_{D}=1 \times 10^{-6} \times 5 \times 10^{28}$ **atoms** $/ \mathrm{m}^{3}$

$n_{e}=5 \times 10^{22} / \mathrm{m}^{3}$

The minority carriers (hole) is $n_{h}=\frac{n_{i}^{2}}{n_{e}}=\frac{\left(1.5 \times 10^{16}\right)^{2}}{5 \times 10^{22}}=\frac{2.25 \times 10^{32}}{5 \times 10^{22}}$

$n_{h}=0.45 \times 10 / \mathrm{m}^{3}$

Similarly, when Boron is implanted a ' p **'type is created with holes** $n_{h}=N_{A}=200 \times 10^{-6} \times 5 \times 10^{28}=1 \times 10^{25} / \mathrm{m}^{3}$

This is far greater than e^{-} **that existed in '** n **' type wafer on which Boron was diffused. Therefore, minority carriers in created '** p **region**

$n_{e}=\frac{n_{i}^{2}}{n_{h}}=\frac{2.25 \times 10^{32}}{1 \times 10^{25}}=2.25 \times 10^{7} / \mathrm{m}^{3}$

Q.17 Answer:

The energy gap in germanium is more than the energy gap in silicon.

Solution:

Silicon is cheaper than germanium, so it is preferred over germanium. But energy gap in germanium is smaller than silicon.

Q.18 Answer:

a semiconductor at $0 \mathrm{~K}$ is zero

Solution:

At $0 \mathrm{~K}$**, the motion of electrons cease and so electric current becomes zero.**

Q.19 Answer:

0.02

Solution:

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The current will flow through diode D_1 and so

$i = \frac{6}{(150+50+100)} = 0.02A$

Q.20 Answer:

$1.01\ \mathrm{mA}$

Solution:

$100 = \frac{\Delta I_c}{\Delta I_b} \Rightarrow \Delta I_b = \frac{1}{100}\ \mathrm{mA} = 0.01\ \mathrm{mA}$

$\Delta I_e = \Delta I_c + \Delta I_b = 1 + 0.01 = 1.01\ \mathrm{mA}$

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